Novia One.

FMP · MODULE 5

OCCUPATIONAL CERTIFICATE · FINANCIAL MARKETS PRACTITIONER

Interest Bearing Securities, at a glance.

A condensed, exam-ready reference for the whole module — all five chapters plus the **maths that lives only in the class deck**. Everything here comes from Novia's own study guide and slides, reduced to the definitions, formulas, directions and traps you are most likely to be tested on.

Money market — the market for **short-term debt securities**, maturities of **less than one year**. Borrowing and lending typically runs **1 day to 180 days**. Deals are called **"open market transactions"** because of their *impersonal and competitive* nature — the price is open, the identity of the counterparty need not be known.

EXAM WATCH The **most important economic role** of the money market is to provide an efficient means for economic units to **adjust their liquidity positions**. Learn that phrase verbatim.

Three defining characteristics

- **Low default risk**
- **Short-term maturity**
- **High marketability**

A **wholesale** market in very large amounts — the **most active financial market by trading volume**. **SARB** uses it to implement monetary policy and manage money in circulation.

Functions performed

- Fund raising
- Cash management
- Risk management
- Speculation or position financing
- Signalling
- Access to information on prices

TRAP A long-term bond is **reclassified as a money market instrument** once its **remaining** maturity falls under one year. Instrument class follows **residual** maturity, not original maturity.

Characteristics of the market

Short-term maturities · High liquidity · Negligible risk · Bulk transactions · Dealer-oriented · Central bank policies · Financing trade · Call money

Call money = short-term borrowing/lending where funds are placed with a financial institution **without a fixed maturity date**. Instruments trade **OTC**, and an **electronic trading system exists in South Africa**.

Participants — investors

Insurance companies · hedge funds · pension funds · banks · corporations · individuals · money market funds. **Individuals play a limited role**, investing **indirectly through money market funds**.

Participants — issuers

Corporations · banks · government · **SARB**.

MONEY MARKET VS CAPITAL MARKET — PRIME MCQ MATERIAL

BASIS	MONEY MARKET	CAPITAL MARKET
Purpose	Meet **working capital** requirements	Become part of the **asset base** of the firm
Function	Short-term credit	Long-term credit
Nature	**Informal**	**Formal and regulated**
Classification	**No subdivision**	Primary **and** secondary market
Instruments	T-bills, commercial paper, CDs	**Bonds and stocks**
Mode of transaction	**Over the counter**	**Exchange**
Liquidity	More liquid	Less liquid
Maturity	**1 day to 1 year**	**No stipulated time**
Risk	**Negligible**	**High**
Returns	**Stable**	**Market linked**
Participants	Banks and financial institutions	Stockbrokers, retail investors, insurers, exchanges, underwriters
Relevance to the economy	Helps **increase liquidity**	Helps **mobilise savings**

Advantages

- Lower risk and shorter duration than equities
- Pays more than savings / current accounts
- Some funds are **tax-free**
- Very liquid — easy to sell at fair market value
- Very low fees; stability and security

Disadvantages

- **Purchasing power eroded by inflation**
- Riskier if invested outside government securities
- **Opportunity cost** of better-returning investments
- Can be expensive relative to the low interest earned

MONEY MARKET INSTRUMENTS

INSTRUMENT	KEY IDEA	WATCH-OUTS
Treasury bills	ZAR-denominated, **sold at a discount to par with NO coupon** → in effect **zero-coupon**, redeemed at par. Typical life **4 weeks to 12 months**.	T-bill rates are the **benchmark default-free** rates. A new issue with the **same maturity date** as an existing issue is a **new tranche**.
Commercial paper	Issued **only by large, well-known, creditworthy companies** and **typically unsecured**. Funds liquidity, inventory, receivables.	Though unsecured it is **backed by a bank line of credit**. Financial companies issue **regularly**, non-financial companies **irregularly**. A high rating means a **smaller discount**.
Negotiable certificates of deposit	States a deposit has been made with a bank for a **fixed period**, repaid with interest. A short-term funding source for banks.	**Issued CDs are liabilities; held CDs are assets.** The depositor's advantage is that it is **tradable**; the bank's is fixed-period funds at a **slightly lower price** than a time deposit.
Repurchase agreements	Buy securities with an agreement to **repurchase at a specified date and price** — a **fully collateralised loan**, in essence a loan backed by securities.	**Most repos use government securities.** Dealers and repo brokers earn **commission**; a **direct repo** is done wherever a counterparty can be found.

EXAM WATCH The instrument list itself is examinable: T-bills and other short-term government securities · **interbank loans/deposits** and other bank liabilities · **repos** and similar collateralised short-term loans · **commercial paper** (issued by **non-deposit** entities) · certificates of deposit · interest rate and currency **derivatives**. Investors treat them as **close substitutes**, which is why money market rates are **highly correlated**.

Purpose & structure of the market

The money market transfers short-term funds from **surplus agents** — corporates, financial institutions, individuals, government — to those who need short-term funds. It also serves **public policy objectives**: financing public sector deficits and managing accumulated government deficits.

Debt vs equity — the distinction everything hangs off

Debt

Fixed term for repayment of principal **plus an agreed interest schedule**, so a **fixed rate of return (YTM) can be calculated** in advance. Selling before maturity may realise a **capital gain or loss**. Carries **less risk** than equity.

Equity

No fixed term and **no guarantee of dividends** — they are paid at the company's **discretion**. Therefore equities offer **no specified rate of return**.

Fixed income offers a **regular income stream**, an element of **capital protection** and some **capital growth** — a **less risky alternative to owning shares**. **Bonds are the most common** fixed-income security; Treasury-issued paper is backed by the credit of the SA government, so **very low risk but low return**.

Features of a debt security

Issue date · issue price · **redemption date** · **coupon rate** · **yield to maturity** · maturity (**short-term ≤ 1 year, long-term > 1 year**) · **collateral** (secured or unsecured) · contractual rights.

Price–yield relationship — inverse. Once issued the coupon is **fixed**. If rates fall, new bonds offer a lower coupon → existing higher-coupon bonds become **more valuable** → **price rises** → because the bond now costs more, the return an investor earns on it **falls, so yield falls**.

EXAM WATCH Learn the **causal chain**, not just the word “inverse”. Novia's distractors reverse one link in the middle.

Who is in the bond market

Issuers — sell bonds to raise money, mostly governments, banks and corporates · **underwriters** — investment banks and firms that help issuers sell the bonds · **bond purchasers** — corporations, governments and individuals buying the debt.

THE YIELD CURVE

The yield curve is the **term structure of interest rates** — the yield on bonds across **terms to maturity**. The government bond curve is the **risk-free yield curve**, "risk free" because governments are not expected to fail to repay borrowing **issued in their own currency**. The **policy interest rate** forms the **beginning of the curve**, being the shortest-term (**overnight**) rate in the economy.

SHAPE	SHORT VS LONG	WHAT IT SIGNALS
Normal	Short-term **lower** than long-term	Slopes **upward**; the ascending slope reflects the **liquidity premium**
Inverted	Short-term **higher** than long-term	Slopes **downward**; investors expect the **future policy rate to be lower**
Flat	Short-term **equal** to long-term	Often a **transition** between normal and inverted

Why the curve matters

- Central to the **transmission of monetary policy**
- A source of information on investors' expectations for **future rates, growth and inflation**
- A determinant of the **profitability of banks**

What makes it change

- Changes in **monetary policy**
- Changes in perceptions of risk — **credit, liquidity, term/duration**
- Changes in the **demand for or supply of bonds**

CORPORATE VS GOVERNMENT Corporate issuers generally issue at a **higher yield than government** because they are riskier. A failure to pay is a **"default"**.

EXAM WATCH An **inverted** curve is not simply "short rates are high" — the material's reason is that investors think the **future policy rate will be LOWER than the current policy rate**. A **flat** curve is described as **transitioning** between normal and inverted, so "the economy is stable" is the wrong reading here.

Bond indenture — the legal contract between **issuer (borrower)** and **bondholders (lenders)**. In South Africa it is called the **trust deed**; in the US the **bond indenture**. It defines the obligations of and restrictions on the borrower, and its provisions are **covenants**.

Negative covenants = prohibitions

- Restrictions on **asset sales**
- **Negative pledge** of collateral — the same assets cannot back several debt issues simultaneously
- Restrictions on **additional borrowings**

They protect bondholders by preventing actions that raise default risk — but must not be so restrictive that the firm cannot respond to opportunities or changing circumstances.

Affirmative covenants = promises to perform

- Make **timely interest and principal payments**
- **Insure and maintain** the assets
- **Comply** with applicable laws and regulations

They **do not typically restrict operating decisions**.

Bond maturity

The date the principal is repaid; the time remaining is the **term to maturity** or **tenor**. On issue, terms run **one day to 30 years**. At maturity the debt **ceases to exist** (redemption day).

1–5 yrs · short | 5–12 yrs · medium | 12 yrs+ · long | No maturity date = **perpetual bond**

Perpetual bonds pay periodic interest but carry **no promise to repay principal**. Maturity matters because it reflects the period over which the lender expects payments (**liquidity needs**), and because **longer maturity** → **higher yield** and **greater expected price volatility**.

Duration

A measure of **how sensitive a bond's price is to interest rate movements** — the **weighted average of the present value of the bond's cash flows**. Expressed **in years** and **typically less than the bond's maturity**. Affected by the **size of the coupon payments** and the **face value**.

- Approximate price change for a **100 basis point** (1 percentage point) rate move: a **2-year duration** bond moves **2%**, a **five-year duration** bond moves **5%**.
- A **weighted average duration** can be computed for a whole portfolio.
- **For a zero-coupon bond, duration = maturity** — no coupons, so the single cash flow arrives at maturity. Zeros therefore give the **most price movement** per rate change, which makes them attractive to investors expecting rates to **decline**.

SCOPE The study guide **defers the duration calculation to Module 7**. Expect Module 5 questions on duration **concepts and direction**, not arithmetic.

The role of bonds in a portfolio

- **Capital preservation** — principal is repaid at a specified maturity date; suits investors who cannot risk capital and those meeting a **liability at a particular future time**.
- **Income** — most bonds pay “fixed” income on a set schedule and issuers are **obligated** to pay the coupons, whereas dividends are smaller and **discretionary**.
- **Capital appreciation** — from a **fall in interest rates** or an **improvement in the issuer's credit standing**. Held to maturity the gain is **not realised**: the price **reverts to par (100)**. **Total return = income + capital appreciation.**
- **Diversification** — reduces the risk of low or negative portfolio returns.
- **Hedge against slowdown or deflation** — inflation makes fixed income less attractive; slower growth → lower inflation → that income becomes more attractive. In **deflation** it is worth even more, so demand, prices and returns rise.

BOND & CREDIT RATINGS

Bond rating — an assessment of the **creditworthiness of the bond's issuer**; a **prediction of the likelihood of default**. Graded **AAA to D** or **Aaa to C**, varying slightly by agency. A **lower rating requires a higher return** to compensate for the added risk. Ratings are meant to be **independent and objective**.

AGENCY	INVESTMENT GRADE — LOW RISK, LOW RETURN	JUNK / HIGH YIELD — HIGH RISK, HIGH RETURN
Moody's	Aaa · Aa1 Aa2 Aa3 · A1 A2 A3 · Baa1 Baa2 **Baa3**	Ba1 Ba2 Ba3 · B1 B2 B3 · Caa1 Caa2 Caa3
S&P / Fitch	AAA · AA+ AA AA− · A+ A A− · BBB+ BBB **BBB−**	BB+ BB BB− · B+ B B− · CCC+ CCC CCC− · D
DBRS	AAA · AA(high) AA AA(low) · A(high) A A(low) · BBB(high) BBB **BBB(low)**	BB(high) ... CCC(low) · D

EXAM WATCH The **investment-grade / junk boundary sits between Baa3 / BBB− (above) and Ba1 / BB+ (below)** — the single most likely ratings question. And ratings are **NOT buy/sell/hold recommendations**: they measure only the issuer's **ability and willingness to repay debt**.

Sovereign ratings

A rating can refer to a **specific obligation** or to the **general creditworthiness of the entity** — a **sovereign** rating is the latter, signifying a country's ability to provide a **secure investment environment**. It weighs **economic status · transparency of capital markets · public and private investment flows · foreign direct investment · foreign currency reserves · political stability**. It is the **first metric** most institutional investors check before investing internationally, and most countries strive for **investment grade**.

Agencies & controversies

Stated as **three main agencies** — **Moody's, S&P, Fitch** — though **Morningstar DBRS** is also listed. Ratings cover **short- and long-term** obligations, and in **both local and foreign currency** (a lack of foreign reserves can warrant a lower foreign-currency rating). Since 2008 the criticism is that **issuers pay** for their own ratings; agencies were **disastrously late** downgrading subprime debt; the **2010 Dodd-Frank Act** forced disclosure of rating performance and created liability; in **2013** all three were **sued** over mortgage bonds held in a **Bear Stearns** hedge fund.

BOND TERMINOLOGY YOU MUST BE ABLE TO RECITE

Par value

The principal repaid at maturity. Also **face value, maturity value, redemption value, principal value**. Prices are quoted as a **percentage of par**: a R1,000 par bond quoted at **98** sells for **R980**. Above par = **premium**, below = **discount**, equal = **at par**.

Basis point

One-hundredth of a percentage point (0.01%). A 3% yield rising **12 bp → 3.12%**. A **120 bp** margin = **1.2%**.

Coupon payments

The coupon rate is applied to **par value, not price**: a R1,000 bond with a 6% coupon pays **R60 a year**. Also called the bond's **nominal interest rate**. The **issuer** is the entity borrowing — in SA: government, local governments, agencies such as **Eskom and Transnet**, and corporations.

Current yield vs YTM

Current yield = annual interest ÷ price — pay \$900 for a bond paying \$50/yr → **≈5.6%**. **YTM** = the effective yield if **held to maturity**, combining the interest paid, the price paid and the **discount or premium to par** recovered at maturity.

Instrument variants

VARIANT	WHAT IT DOES
Zero-coupon pure discount bond	Does pay interest, but **not at regular intervals** — it **accumulates and is paid as one lump sum at maturity**, achieved by selling at a **deep discount to par**. A 10-year R1,000 zero yielding 7% sells at about **R600**; the **R400 difference is the bondholder's compensation**.
Convertible	Holder may **convert into common shares** at a certain exchange rate. Because of the equity upside, convertibles **pay lower interest** than comparable non-convertible bonds.
Step-up note	Coupon **increases over time on a predetermined schedule**; typically **callable at each step-up date**. If market yields **rise**, the bond is **less likely to be called**, so the holder may keep the higher coupon — giving **some protection against rising rates**. **Single** = one increase, **multiple** = several.
Floating-rate note	**Coupon = reference rate + quoted margin** — e.g. LIBOR + 0.75%; if 1-year LIBOR is 2.3% the bond pays **3.05%** of par. A **cap benefits the issuer**, a **floor benefits the bondholder**. An **inverse floater**'s coupon **rises when the reference rate falls**. Many pay **quarterly**. SA benchmark = **JIBAR** (1, 3, 6, 12 month); internationally **LIBOR**.

Inflation protection securities (IPS)

A **variant of a floating-rate security** — the only difference is that the reference is the **inflation rate**. The **coupon RATE stays constant**; the **PRINCIPAL is increased** by inflation (decreased by deflation), so the **coupon paid** rises and falls with inflation. **Coupon rate = CPI + xxx basis points.** US **TIPS**: principal adjusted by the **CPI**, interest paid **every six months**, maturities of **5, 10 and 30 years**.

WORKED EXAMPLE R100,000 par IPS trading at par, **4% coupon paid semi-annually**, annual inflation **2.5%**. Semi-annual inflation = 2.5 ÷ 2 = **1.25%** → adjusted principal = 100,000 × 1.0125 = **R101,250** → coupon = (4% ÷ 2) × 101,250 = **R2,025**.
Method: inflation-adjust the principal FIRST, then apply half the stated annual coupon rate.

STRUCTURE	HOW THE PRINCIPAL COMES BACK
Bullet (typical)	Periodic coupons over the life, **principal paid with the final interest payment** at maturity. R1,000 face, 5-year, 5% annual coupon: **50 · 50 · 50 · 50 · 1,050**.
Balloon payment	Where the final payment includes a **lump sum in addition to the final period's interest**.
Fully amortising	Each payment includes **interest and some repayment of principal**; the principal is fully paid off by the last payment. The same bond amortising: **R230.97 × 4, then R230.98**. Typically **car and home loans**.
Sinking fund	Principal repaid at **several dates** rather than one — e.g. an RSA bond paying 10.5% semi-annually with principal in **2025, 2026 and 2027**.

EXAM WATCH The IPS distractor is “the **coupon rate** is adjusted for inflation”. It is not — the **principal** is adjusted and the **rate stays fixed**. The amount of **cash** paid does change, which is what makes the wrong option sound right.

CONTINGENCY PROVISIONS — THE EMBEDDED OPTIONS

Contingency provision — an action that **may** be taken if an event (the contingency) occurs. In bond indentures these are **embedded options**: "embedded" because they are an **integral part of the bond contract, NOT a separate security**. Some are exercisable at the option of the **issuer**, others at the option of the **purchaser**. Bonds **without** them are **straight** or **option-free** bonds. The three are **call · put · conversion privilege**.

OPTION	WHOSE RIGHT	EXERCISED WHEN	EFFECT ON YIELD AND PRICE
Call	**Issuer**	**Interest rates are low**, or the issuer's **credit quality has improved** — it can redeem and reissue at a lower yield	**HIGHER yield / LOWER price** than an identical non-callable bond; the difference equals the **value of the call to the issuer**. The bondholder carries **reinvestment risk**, because bonds are called precisely when proceeds can only be reinvested at a lower yield.
Put	**Bondholder**	Fair value falls below the put price — because **rates have risen** or the **issuer's credit quality has fallen**	**LOWER yield / HIGHER price** than an identical option-free bond — the exact **mirror image** of the callable result, because the choice belongs to the bondholder.
Conversion	**Bondholder**	The share price rises far enough — though holders often wait, because the **bond's interest yield exceeds the dividend yield** on the shares	A **hybrid**: downside protection of a bond plus the upside of equity, so issued at a **lower yield**. The issuer gains a lower interest cost, and **debt financing converts to equity financing**.

Call provision — the worked timeline

A 6% 20-year bond issued at par on **1 June 2012**, redeemable at **102% of par after 1 June 2017**, **101% after 1 June 2020** and **100% after 1 June 2022**:

- The first five years carry **call protection** — also called a **lockout period, a cushion or a deferment period**.
- **1 June 2017** is the **first call date** at a call price of **102**. The excess over par is the **call premium** — here **2%, or R20 per R1,000 bond**.
- After **1 June 2022** the bond is callable **at par**, and that date is the **first par call date**.
- For a currently callable bond the **call price puts an upper limit on the bond's market value**.

Convertible bonds — the three numbers

Conversion price	**price per share at which the bond (at par) may convert**
Conversion ratio	**par ÷ conversion price → 1,000 / 40 = 25 shares**
Conversion value	**ratio × share price → 25 × R50 = R1,250**

Typically issued with maturities of **5–10 years**. Even when the conversion value far exceeds par, holders **may not convert** until forced, because the **interest yield on the bond exceeds the dividend yield** on the shares. That is why many convertibles carry a **call provision**: because the **call price is less than the conversion value**, calling **forces bondholders to exercise the conversion option**.

Investors also **borrow funds to buy bonds** — for which the **returns must exceed the borrowing costs**. The securities purchased are used as **collateral**, and two collateralised arrangements exist.

Margin buying

- Funds are provided by the **broker**, who borrows from the **bank**.
- The broker charges the investor the **call money rate** = the interest the bank charges the broker **+ a service charge**.

Repurchase agreements

- The investor **sells** a security with a commitment to **buy it back** on a specified repurchase date.
- The **difference between the selling and repurchase price is the cost of the loan**, from which the **repo rate** is calculated.
- **Overnight** (one day) or **term** (longer than one day); mostly used by **institutional investors** in the bond market.

WORKED EXAMPLE A firm sells a 4%, 12-year bond — **par R1,000,000**, **market value R970,000** — for **R940,000**, repurchasing it 90 days later for **R947,050**.
· Implicit rate for the 90-day period = **947,050 / 940,000 − 1 = 0.75%**; the repo rate is then expressed as the equivalent annual rate.
· **Repo margin (haircut)** = market value vs amount loaned = **940,000 / 970,000 − 1 = −3.1%**, and the margin **protects the lender** if the security's value falls over the term.

REPO RATE VS REPO MARGIN — KEEP THE TWO LISTS APART

The repo RATE is...

- **Higher** the longer the repo term
- **Lower** the higher the credit quality of the **collateral security**
- **Lower** when the collateral is **delivered to the lender**
- **Higher** when interest rates for **alternative sources of funds** are higher

The repo MARGIN is...

- **Higher** the longer the repo term
- **Lower** the higher the credit quality of the **collateral security**
- **Lower** the higher the credit quality of the **borrower**
- **Lower** when the collateral security is in **high demand or low supply**

EXAM WATCH The two lists **overlap but are not identical**. Only the **margin** list mentions the **borrower's** credit quality and the **demand/supply of the collateral**; only the **rate** list mentions **delivery of the collateral** and **alternative funding rates**. Novia can test either, so keep them separate.

Repos, in one place

A repo is an arrangement where one party **sells a security with a commitment to buy it back at a later date at a specified (higher) price**. The repurchase price exceeds the selling price and accounts for the **interest charged by the buyer**, who is **in effect lending funds to the seller with the security as collateral**. The rate implied by the two prices is the **repo rate** — the **annualised percentage difference** between them. The interest cost of a repo is **customarily LESS than the rate on bank loans** or other short-term borrowing.

Uses of the repo

- Banks invest surplus cash
- Money market returns at **sovereign risk**
- Raising funds by borrowers
- Liquidity adjustment in the system

Central banks conduct short-term repos on bonds of **any residual maturity** to influence money market rates. Participants: **central banks, financial institutions, non-financial corporations**.

The bond market — also called the **debt market, fixed-income market or credit market** — is the collective name for all trades and issues of debt securities. Governments issue to **pay down debt and fund infrastructure**; companies issue to **maintain operations, grow product lines or open new locations**. Bonds are **issued in the primary market** and **traded in the secondary market**.

Primary bond market

- The **issuer sets most of the terms**, including the initial price and interest rate
- Investors **buy at face value**, usually with a fixed coupon and maturity date
- The **issuer receives the proceeds** of the sale

Secondary bond market

- Lets investors sell quickly and **convert into cash before maturity**
- Transactions are between **two investors** — the **issuer is not involved**; a brokerage may act as intermediary or as principal
- Price is set by **market conditions**: prevailing rates, supply and demand, credit quality

The JSE Debt Market

The JSE regulates the **largest listed debt market in Africa** by both market capitalisation and liquidity, and has done since **2009**, when it acquired the **Bond Exchange of South Africa**.

Listed debt instruments (end-2013)	**≈1,600 · more than R1.8 trillion nominal**
Placed by the SA government	**more than half of all listed debt**
Traded daily	**roughly R25 billion**
Government bonds	**>R1 trillion listed · 90% of reported liquidity**
Primary dealers	**12 appointed in 1998 · 8 at end-2013**
Corporate bonds	**first issued 1992 · >1,500 instruments listed**
Repo market daily funding	**>R25 billion · daily spikes >R200 billion**

Bonds listed on the JSE: **Green Bonds · Government Bonds · Corporate Bonds · Repo Bonds**. Primary dealers are required to **make a secondary market in qualifying RSA paper**. Government bond markets are subject to a **high level of supervision by the central bank**, and government bonds often serve as a **benchmark for pricing other assets** and as **collateral**.

Overview of bond sectors

Internal bond market

Domestic bond market — the issuers are **domiciled in the country** and trade the bonds within it.

Foreign bond market — the issuers are **not domiciled** in the country, but their bonds are **traded** in it (e.g. UK bonds traded in the South African bond market).

External bond market

The **offshore** or **Eurobond** market. At issuance the bonds are **offered simultaneously to investors in several countries**, and they are **issued outside the jurisdiction of any single country**.

Government bond markets

Treasury securities are issued **via an auction**. The **most recently auctioned issue for a maturity** is the **on-the-run** (current) issue; the securities it replaces are the **off-the-run** issues.

FIXED PRINCIPLE SECURITY	COUPON	MATURITY
Treasury bills	Issued at a **discount** to par, **no coupon rate**, mature at par	**Less than 12 months**
Treasury notes	**Coupon paying**, issued at par value, mature at par value	More than a year but **less than 10 years**
Treasury bonds	**Coupon paying** and **not callable**	**More than 10 years**

Inflation indexed bonds

Inflation protected bonds — the **coupon and principal are adjusted semi-annually** to reflect inflation. The **coupon rate (the real rate) is fixed**, and the coupon is **calculated on the adjusted principal**. Coupons rise with inflation, so TIPS provide inflation protection — but the **increase is taxable**, making them **not so favourable**. In **deflation** the par value at maturity can be **less than at issuance**.

Treasury STRIPS

The Treasury does **not issue long zero-coupon bonds** — only T-bills. Because zero-coupon bonds with **no credit risk** are in demand, they are **created by stripping the coupon payments** off a Treasury bond. Strips from coupons are **coupon strips**; strips from principal are **principal strips**.

Asset-backed securities

Fixed-income instruments created through **securitisation**: ownership of assets is **transferred from the original owners into a special legal entity**, which then **issues securities backed by those assets**; the **assets' cash flows** pay the interest and repay the principal owed to the holders. The pool of securitised assets from which the cash flows are generated is called the **collateral**.

Typical **securitised assets** are loans and receivables: **residential mortgages, commercial mortgages, auto loans, student loans, bank loans, accounts receivable and credit card receivables** — and, with innovation, all kinds of income-yielding assets including **airport landing slots and toll roads**.

- Gives investors **direct access to liquid investments and payment streams** unattainable if all financing ran through banks.
- Enables banks to **increase loan originations** at greater economic scale than their own loan portfolios allow.
- Contributes to **lower borrowing costs**, **higher risk-adjusted returns** for investors and greater **efficiency and profitability** for the banking sector.

RELATIONSHIP	THE BOND TRADES AT	PRICE VS PAR
Coupon rate = yield (YTM)	**Par**	= 100
Coupon rate > yield	**Premium**	> 100
Coupon rate < yield	**Discount**	< 100

Prior to maturity a bond's price fluctuates on the **issuer's financial stability, rating, coupon, years to maturity, interest rates and the supply and demand** for the security — any of which, separately or together, can move it to a discount, par or a premium.

CONSTANT-YIELD PRICE TRAJECTORY Whether at a premium or a discount, a bond's value **converges to par at maturity** — that convergence is the **constant-yield price trajectory**, showing how the price would move over time if the YTM stayed constant. The guide's 6% bond: at a **3% YTM** it is worth **1,085.458**; at **6%**, **1,000.00**; at **12%**, **852.480** — and all three arrive at **1,000 at maturity**.

FACTORS AFFECTING BOND YIELDS

Bonds are traded **on a yield basis in South Africa**, and the **yield determines the price**. The guide gives five determinants.

1 · Market interest rate levels

The rate on a particular bond **cannot be out of line** with rates on other financial assets. Market rate levels are determined by **inflation expectations, exchange rate considerations and tax policies**; when those change they affect the prices and yields of **all** financial instruments in a similar way.

2 · Maturity

The relationship between maturity and yield is the **term structure of interest rates**, drawn as the **yield curve**. An upward slope means the **longer the term to maturity, the higher the yield** — and that ascending slope **reflects the liquidity premium**. Investors require a premium to hold longer-dated bonds; issuers pay it because the **uncertainty of rolling debt over in future is reduced**. Curves are not always upward — falling, humped or flat shapes show that factors **other than liquidity** (expectations and market forces) also drive yields.

3 · Future interest rates (expectations)

When rates are **expected to rise**, investors will not buy long-term bonds now — they expect to buy them at a **higher rate (lower price)** soon — so sellers of long bonds must offer a premium, causing an **upward-sloping curve**. When rates are **expected to fall**, investors prefer long bonds, so sellers of **short** bonds must offer the premium, giving a **downward-sloping curve**. An expectation of higher rates can also push floating-rate holders to fix via the **swap market**, forcing market-makers to **sell fixed-rate bonds as a hedge** — more supply, higher rates.

4 · Market forces

Demand and supply for **specific** bonds: increased demand for short-term bonds **raises their price and lowers their yield**. Investors have segment preferences — short-term insurers versus long-horizon funds — so their activity affects prices and yields **within that market segment**. Note the SA observation: the number of **new government bond issues has decreased**, and that reduced supply has at times put **downward pressure on bond yields**, giving **more scope for corporate debt issues**.

5 · Issuer

The issuer drives the risk attaching to the bond. **The higher the quality of the borrower or issuer, the lower the risk of default**, so the investor can expect a **lower risk premium or yield**. Conversely, **borrowers of lower credit quality must pay a higher rate of interest** to compensate investors for the additional risk they bear.

Three sources of return from a fixed-rate bond

- **Coupon and principal payments**
- **Interest earned on coupon payments reinvested** over the investor's holding period
- Any **capital gain or loss** if the bond is sold prior to maturity

RISK	WHAT IT IS · HOW IT BEHAVES
Interest rate risk	Losses triggered by an **upward move in prevailing rates** for new debt instruments. **Rates up → prices down** (investors buy the new higher-yielding bonds, making older ones less valuable); **rates down → prices up**. Reduce it by buying bonds of **different durations** or by hedging with **interest rate swaps, options or other derivatives**.
Yield curve risk	The general assumption that the curve shifts in a **parallel** fashion **does not hold for all maturities**. Different maturities react differently, so every bond in a portfolio moves a **different number of basis points**. That differing exposure is **yield curve risk**, measured with **key rate duration** — the duration of key assets holding the others constant.
Reinvestment risk	Being unable to reinvest cash flows (e.g. coupons) at a rate comparable to the current return. **Longer maturity → higher** reinvestment risk; **higher coupon → higher** reinvestment risk, so a **premium bond depends more on reinvestment income** than a par bond. **Zero-coupon bonds have NO reinvestment risk if held to maturity.** For long-term bonds in high-rate environments, reinvestment income can be as much as **70% of total dollar return**.
Credit risk	Three components — **default risk** (the issuer fails to pay interest or principal on time); **credit spread risk** (the spread over risk-free widens; spreads **widen in a declining economy and narrow during expansion**); and **downgrade risk** (deterioration in the credit rating — an anticipated downgrade **increases the spread**, which **decreases the bond price**).
Liquidity risk	The bond is sold **below its indicated value** for lack of demand. It is measured by the **bid–ask spread** — the **wider the spread, the higher the liquidity risk** — and changes over time with the demand and supply of the issue.
Inflation / purchasing power risk	The bond's total return fails to outpace inflation; **bonds are affected severely**. **Floating-rate bonds give partial protection** (the coupon adjusts, the principal does not); **inflation-linked bonds give full protection** because the **principal on which the coupon is based does adjust**.
Volatility risk	Loss of value from a **change in volatility** — it exists only where the bond has an **embedded option**. As volatility rises, the **option's value rises**; for a **callable** bond a more valuable call means a **less valuable bond**.

Duration in Chapter 4's words

Duration measures **how long it takes, in years, for an investor to be repaid a bond's price through its total cash flows** — and is also used to measure the **sensitivity of a bond's price to changes in interest rates**. Generally, **the higher a bond's duration, the more its price will fall when rates rise**. The two factors that affect it are **time to maturity** and **coupon rate**. A portfolio's duration is the **weighted average of the individual bond durations** held in it.

Macaulay duration

The **weighted average time until all the bond's cash flows are paid**. By accounting for the **present value** of future payments it lets you evaluate and compare bonds **independent of their term**.

Modified duration

Not measured in years. It measures the **expected change in a bond's price given a 1% change in interest rates**.

EXAM WATCH — THE GUIDE AND THE DECK DISAGREE The **study guide** says a bond's **term** is the linear measure and that **duration is nonlinear**, decreasing as time to maturity shortens. The **class deck** calls duration **"a linear measure"**, with the curvature picked up by **convexity**. Both statements are in Novia's own material — read the stem carefully and answer from the source it is quoting.

RATE	MEANING	WORKED FIGURE
Nominal	The **stated** rate — the actual monetary price borrowers pay lenders, **unadjusted for inflation**. Also called the **coupon rate**, because it was traditionally stamped on the coupons bondholders redeemed.	A 5% nominal loan costs **$5 of interest per $100** borrowed.
Real	Factors **inflation** into the equation, giving a more accurate measure of the investor's **buying power** after the position is redeemed.	6% nominal − 4% inflation = **2% real**. **Real rates can be negative.**
Effective	Takes **compounding** into account — a bond may pay interest annually but compound semi-annually, increasing the overall return.	R1,000 at 6% compounding half-yearly: R30 + R30.90 = **R60.90** → effective **6.09%** vs nominal 6%.

RATES IN PRACTICE

Where each rate is used

Nominal rates are what financial institutions advertise on loans and deposits, and what mortgages, personal loans and credit cards specify. **Real** rates are used to analyse **investment decisions** — the actual purchasing power of a return — for long-term goals such as retirement, and when comparing international investments across different macroeconomic policies.

Who likes effective rates

Borrowers dislike effective rates, because they reflect a **higher cost**. **Savers like** them, because they **earn more with more compounding periods**.

CURVE SHAPES AS ECONOMIC SIGNALS A **normal** yield curve implies **stable economic conditions** and a normal economic cycle. A **steep** curve implies **strong economic growth**, with conditions often accompanied by **higher inflation and higher interest rates**.

CHAPTER 5 · UNDERSTANDING YIELD SPREAD

The interest rate charged on a bond depends on the **risk-free rate** and the **risk associated with the issuer**, and that rate is then affected by the **economy, central bank decisions and government fiscal policies**.

Interest rate policies — the central bank's three tools

TOOL	MECHANISM
Open market operations	Buying and selling **Treasury securities** affects the **money supply** and so **indirectly** affects market interest rates. **Selling** securities **reduces** money supply → rates **increase**; **buying** securities **increases** money supply → rates **decrease**.
Discount rate	The rate at which banks can borrow and lend each other funds on a **collateralised basis**. **Increasing** it increases the cost of borrowing → less borrowing → **reduces the money supply** → market rates **increase**.
Bank reserves	**Increasing the reserve requirements** of banks leaves them less money to extend loans → **reduces the money supply** → interest rates **increase**. The opposite is also true.

Treasury yield curve

Treasury securities **don't expose investors to credit risk**, so they are used as the **minimum required return for risky securities**. The relationship between Treasury yield and maturity plotted graphically is the **Treasury yield curve**, and it is **constructed using on-the-run issues**.

Yield spread

The **difference between a bond's yield and the yield on a reference bond**. Spreads indicate the market's **perception of the risk of a bond relative to the reference bond**. **The less liquidity a bond has, the higher its yield spread relative to Treasuries** — investors require compensation for **giving up liquidity**.

Nominal spread

The difference in yield between the **YTM of a bond** and the **YTM of a comparable benchmark** — for example a 10-year corporate against the 10-year Treasury. A **widening spread** may indicate that the **company is showing signs of weakness**, that the **economy is showing signs of weakness**, **or** that the bond is now **undervalued and a buying opportunity is at hand**. All three readings are offered in the material — watch for a “which is NOT” stem.

CURVE SHAPE	CHAPTER 5'S READING
Normal upward sloping	The **longer the maturity the higher the yield**; yields on longer-term bonds may **continue to rise**, responding to periods of **economic expansion**. Investors expecting even higher long yields **park funds in shorter-term securities**, pushing short yields **even lower** and setting in motion a **steeper** normal curve.
Inverted downward sloping	Also termed the **abnormal** curve: the **longer the maturity the lower the yield**, corresponding to periods of **economic recession**. Investors expecting lower long yields **buy long bonds to lock yields in** → long prices up and long yields down, short prices down and short yields up, **inverting the curve further**.
Flat horizontal	The rate of interest is assumed to be **the same for all maturities**. It **may arise from either** the normal or the inverted curve as conditions change: expansion into slowdown flattens a normal curve; recession into recovery tilts an inverted curve toward flat.

THEORIES OF THE TERM STRUCTURE

Interest rates are **both a barometer of the economy and an instrument for its control**, and the term structure — market interest rates at various maturities — is a **vital input into the valuation of many financial products**. What does the shape of the curve reveal? Three theories answer that.

Pure expectations theory

Any long-term rate simply represents the **geometric mean of current and future one-year interest rates expected to prevail** over the maturity of the issue.

- **Rising curve** → investors expect **short-term rates to rise** in the long term.
- **Flat** → investors expect **no changes**; short- and long-term yields are equal.
- **Inverted** → short-term rates are **higher now but expected to decrease** in future.
- **Humped** → the short-term rate will **rise for a certain period and then fall**.

Liquidity preference theory

Long-term securities **should provide higher returns** than short-term obligations, because investors are **willing to sacrifice some yield** to invest in short maturities and **avoid the higher price volatility of long-maturity bonds**. The longer the maturity the greater the volatility, so a **premium must be added** to compensate for the increased risk — the result is an **upward-sloping curve**.

Also called the **biased expectations theory**, because the curve is determined by **two** factors: the investor's **expectations** and a **liquidity premium**.

Market segmentation theory

- **Demand and supply determine the interest rate for each maturity**, so **any** shape of curve is possible — each market sector is **independent** in determining its rates.
- **Preferred habitat theory** is the variant: investors invest **according to the nature of their liabilities**, and require a **premium to be induced out of their preferred investment sector**.
- Also a **biased** theory.

EXAM WATCH Three-option questions here usually swap the **mechanism**, not the name: “geometric mean of expected future one-year rates” belongs to **pure expectations**; “investors sacrifice yield to avoid long-bond volatility” belongs to **liquidity preference**; “each maturity sector prices independently” belongs to **market segmentation**.

APPENDIX · THE MATHS — CLASS DECK ONLY

SCOPE WARNING The **study guide contains none** of the money-market yield formulas, the bond-pricing formula, the yield-quote conventions or the duration and convexity measures. All of it comes from the **class presentation**, so revise this appendix from the slides.

Discount yield

The annualised rate of return calculated on the **difference between the face value and the purchase price** — the purchase price is typically **less than** face value. **Used for T-bills** and other short-term securities where the interest is effectively the face-value/purchase-price gap.

Measured against **face value**.

Add-on yield

The annualised yield based on the **total interest payment over the term divided by the principal amount invested**. **Used for consumer loans**, where interest is calculated and **added onto the principal upfront** and the borrower repays principal plus interest over the term.

Measured against **principal invested**.

Both are quoted as **simple annual interest**. The choice between them depends on the **nature of the instrument** and the **conventions or requirements of the industry or regulator**. **The discriminator is the base** the yield is measured against: same cash flows, different quoted number.

BOND VALUATION — THE PRESENT VALUE MODEL

The **intrinsic value of a bond is the present value of all its future cash flows**, discounted at the required rate of return. For a **plain vanilla bond** — a straight, option-free bond paying a fixed coupon and repaying par at maturity — that is written as:

$$\text{Bond Value} = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{F}{(1+r)^n}$$

C	**coupon payment**
F	**face value (par value)**
r	**discount rate (market yield)**
n	**number of periods until maturity**

In words: **the present value of the coupon stream (an annuity) plus the present value of the par value (a single lump sum)**. Closed form for hand calculation: $\mathbf{P = C \times [1 - (1 + r)^{-n}] \div r \; + \; F \times (1 + r)^{-n}}$.

The same equation as the class deck writes it

The deck states it for a **semi-annual** payer, which is why the coupon and the rate are halved and the periods doubled:

$$P_m = \sum_{t=1}^{2n} \frac{C_t / 2}{(1+i/2)^t} + \frac{P_p}{(1+i/2)^{2n}}$$

P_m	**current market price / value of the bond today** (= Bond Value)
C_t	**the annual coupon payment** (= C)
i	**the prevailing yield to maturity** (= r)
n	**number of years to maturity**
P_p	**par value** (= F)

They are the **same model** — only the **period** differs. If the bond pays **annually**, use **C, r and n** exactly as given; if it pays **semi-annually**, halve the coupon, halve the rate and double the periods.

EXAM WATCH **Semi-annual bond → halve the coupon AND the rate, and double the number of periods (2n).** That is the single most common arithmetic slip, and the slide flags it explicitly. The deck's own advice is to use a **financial calculator — PV, FV, N, I/Y, PMT** — since you will normally be given everything and asked to solve for one.

Accrued interest, clean price and full price

- **Full price** (invoice price or **dirty** price) = the price **including accrued interest** at settlement.
- **Flat price** (clean price or **quoted** price) = **full price − accrued interest**.
- So **accrued interest + clean price = full (dirty) price**.
- **Accrued interest is paid to the SELLER** — it is the **proportional share of the upcoming interest payment**, earned from the **last coupon date to the sale date**. Including it in the full price means the seller does not lose the interest earned for the part of the accrual period during which they owned the bond.

The price–yield curve

The relationship between the **coupon rate, the yield and the price** resolves to exactly the **premium / par / discount** rule: coupon **=** yield → **par**; coupon **>** yield → **premium**; coupon **<** yield → **discount**. The curve is **not a straight line** — that curvature is what **convexity** measures.

HOW TO USE THIS APPENDIX Nothing here appears in the study guide, so if a question involves a **calculation**, a **quote convention** or a **duration measure**, it is being drawn from the deck — and the deck's wording is what the answer options will mirror.

YIELD MEASURES

MEASURE	WHAT IT MEASURES
Current yield	The income return on a bond — **annual coupon ÷ current bond price**
Yield to maturity (YTM)	The return **if held to maturity** and all coupons are **reinvested at the YTM**
Nominal yield	The percentage of interest paid on the bond periodically — **annual coupon ÷ PAR value**
Yield to call (YTC)	The yield on a **callable** bond repaid or called by the issuer before maturity — which **usually happens when interest rates are decreasing**
Realized / holding-period yield	The yield where the investor **sells prior to maturity** in the secondary market
Effective annual yield (EAY)	The annual yield with **compounding** taken into account
Bond equivalent yield (BEY)	The **nominal annual rate**

EXAM WATCH **Nominal yield uses PAR as the denominator; current yield uses PRICE.** That is the whole distinction between them — and a natural three-option MCQ.

Novia's SA / USA quote convention

All bond prices can be quoted on **either yield or price**. The deck teaches:

- **South Africa quotes on YIELD** — "**Yield Quotes are always interpreted as a 100 − YTM.**"
  Example: a **R5,000 par municipal bond quoted at 7.25%** → price = **100 − 7.25 = 92.75**, i.e. roughly **92.75% of par**.
- **The USA quotes on PRICE** — "**Price Quotes are always interpreted as a percentage of par.**"
  Example: a quote of **98½** means **98.5% of par**, not $98.50. On **$5,000** par → 5,000 × 98.5% = **$4,925**. Equivalently 100 − 98.5 = 1.5% → $5,000 − (1.5% × $5,000) = **$4,925**.

THE QUOTE QUIRK — READ THIS TWICE

HIGHEST-RISK ITEM IN THE MODULE The "**price = 100 − YTM**" rule is **not standard bond mathematics** — a 7.25% yield does not genuinely imply a price of 92.75% of par. But it **is** what Novia teaches and examines, complete with a worked example. **Answer per the guide and deck**, and raise the objection only if you are asked to justify. This is the single most dangerous item in Module 5 for anyone answering from real-world fixed-income knowledge.

Reading a bond quote

`RATE 3 | MO/YR Feb 09 | BID 100:14 | ASKED 100:15 | CHG +8 | ASK YLD 2.22`

- A **3 percent** Treasury note **due February 2009**.
- **Bid 100:14** means **100 + 14/32 = 100.4375% of par** — the **colon denotes 32nds**, and the same applies to the asked price.
- Buying at the **asked** price gives a **YTM of 2.22%**.

PRICE VOLATILITY

High price volatility — high interest rate sensitivity — means the bond experiences a **relatively large percentage price change for a given change in yields**, which can significantly affect the value of a portfolio.

Determinants of volatility

- **Longer** time to maturity → **higher** sensitivity
- **Lower** coupon rate → **higher** sensitivity
- **Lower** yield to maturity → **higher** sensitivity

Mnemonic: **long, low and low = most volatile** — exactly why a **zero-coupon** bond, having the lowest possible coupon, is the most rate-sensitive bond of a given maturity.

Factors affecting duration

- **Longer** maturity → **higher** duration (except for some discount bonds)
- **Higher** coupon rate → **lower** duration
- **Higher** YTM → **lower** duration

These run **parallel** to the volatility determinants — higher duration means higher sensitivity. Note the parenthetical caveat: an absolute-sounding option may be the trap.

DURATION & CONVEXITY — THE DECK'S TREATMENT

Duration

A **linear** measure of the sensitivity of a bond's price to fluctuations in interest rates: the **weighted average time to full recovery of the principal and interest payments**. It is measured **in units of time** and is **always less than or equal to the bond's maturity**, because the value of more distant cash flows is more sensitive to the interest rate.

The critical risks affecting a bond are **interest rate risk** and **term to maturity** — so **to manage interest rate risk, you manage duration**. Duration incorporates the bond's **yield, coupon, maturity and call features**.

Macaulay duration	**weighted average term to maturity of the cash flows**
Modified duration	**an approximation of the bond's rate sensitivity**
Effective duration	**a direct measure of rate sensitivity, for any instrument**

Convexity

Duration is a **linear** measure, but the true price–yield relationship is **curved**. **Convexity captures that curvature** — the **second-order** effect that duration alone misses. It is what explains why duration **under-estimates the price rise when yields fall** and **over-estimates the price fall when yields rise**.

EXAM WATCH If an option says duration measures price sensitivity **exactly**, it is wrong — duration is an **approximation**, and convexity is the correction. Conversely, if an option says convexity measures **time**, it is wrong — only duration is measured in units of time.

THE NIGHT-BEFORE LIST · MONEY MARKETS TO RATINGS

1. **Money market:** maturities **under one year**, lending **1 day to 180 days**, "**open market transactions**", and the most important role is **adjusting liquidity positions**.
2. **Money vs capital market:** informal vs **formal and regulated** · **OTC** vs **exchange** · no subdivision vs **primary + secondary** · negligible vs **high** risk · stable vs **market-linked** returns · increases liquidity vs **mobilises savings**.
3. **T-bills:** sold at a **discount**, **no coupon**, zero-coupon in effect, redeemed at par, the benchmark **default-free** rate.
4. **Commercial paper: unsecured**, issued only by large creditworthy companies, **backed by a bank line of credit**.
5. **NCDs: issued CDs are liabilities, held CDs are assets.**
6. **Repo = fully collateralised loan**; the **margin (haircut) protects the lender**; keep the rate and margin direction lists separate.
7. **Price–yield is inverse** — and know the causal chain, not just the label.
8. **Curve shapes:** normal · **inverted** (short > long, expecting a **lower future policy rate**) · flat (**transitioning**). The government curve is the **risk-free curve**, and the **policy rate starts the curve**.
9. **Covenants:** negative = **prohibitions**, affirmative = **promises to perform**.
10. **Maturity bands:** 1–5 short · 5–12 medium · 12+ long; **no maturity date = perpetual**.
11. **Zero-coupon: duration = maturity** → the most price movement per rate change, and **no reinvestment risk** if held to maturity.
12. **Investment grade / junk boundary: Baa3 / BBB− above, Ba1 / BB+ below.**
13. **Ratings are NOT buy/sell/hold recommendations** — they measure only **ability and willingness to repay**.

14. **Callable → higher yield / lower price** (value to the **issuer**); **putable → lower yield / higher price** (value to the **bondholder**). Mirror images — the direction is the question.
15. **Convertible: conversion ratio = par ÷ conversion price; conversion value = ratio × share price.** A hybrid at a lower yield, and a **call forces conversion**.
16. **IPS: the coupon RATE is constant, the PRINCIPAL is inflation-adjusted** → R100,000 at 4% semi-annual with 2.5% inflation pays **R2,025**.
17. **Bullet vs fully amortising vs sinking fund**; a **balloon** is a lump sum on top of the final interest payment.
18. **Premium / par / discount** follows **coupon vs yield**, and every bond **converges to par at maturity**.
19. **T-bills < 12 months · T-notes 1–10 years · T-bonds 10 years+ and not callable**; **on-the-run** = most recently auctioned.
20. **Nominal yield ÷ PAR** vs **current yield ÷ PRICE**.
21. **Semi-annual pricing:** halve the coupon **and** the rate, and double the periods.
22. **Accrued interest + clean price = dirty price**; accrued interest is **paid to the seller**.
23. **Volatility and duration both rise with: longer maturity, lower coupon, lower YTM.**
24. **Credit risk = default + credit spread + downgrade risk**; spreads **widen in a declining economy**.
25. **Liquidity risk is measured by the bid–ask spread**; less liquidity → a **higher spread over Treasuries**.
26. **Reinvestment risk rises with longer maturity and a higher coupon**; zeros held to maturity have none.
27. **Pure expectations** (geometric mean of expected future one-year rates) vs **liquidity preference** (a premium for long-bond volatility) vs **market segmentation** (independent sectors; preferred habitat).
28. ⚠️ **SA yield quote = 100 − YTM; USA price quote = % of par** (a colon means **32nds**). Answer per Novia.

KNOWN ERRORS IN NOVIA'S OWN MODULE 5 MATERIAL

Worth knowing so a mis-typed stem doesn't throw you in the exam.

WHERE	WHAT IT SAYS	WHAT IT SHOULD SAY
Guide p.25 IPS worked example	"(1 + 0.0.12) x 1000 = 101 250"	**(1 + 0.0125) × 100,000 = 101,250.** The final answer **R2,025 is correct**; the printed working is not.
Guide p.42	"the **structure** of interest rates"	The **term structure** of interest rates.
Guide §2.6 vs §2.6.2	"one of **three** credit rating agencies", then **four** are listed (Moody's, S&P, Fitch, Morningstar DBRS)	If asked **how many**, the guide's stated number is **three**.
Guide p.14	"the **cash rate in Australia**" as the policy rate	Borrowed RBA source text. The SA equivalent is the **repo rate**.
Guide Ch5 title	"Understanding **Yielod** Spread"	**Yield** spread — a typo in the chapter heading itself.
Deck slides 79 / 81	"Yield quotes are always interpreted as a **100 – YTM**"	Not standard bond mathematics — but **examinable**. Answer their way.
Guide §4.2 vs the deck	Guide: **duration is nonlinear**. Deck: duration is **a linear measure**.	Both appear in Novia's material. Answer from whichever source the question quotes.

SOURCES **FMP V05 Mod05 Int Bearing Securities & Bonds STUDY GUIDE 25 08 21 CJ** (47pp, five chapters) and **FMP V06 Mod05 Int Bearing Securities & Bonds CLASS Pres 26 08 04 CJ** (124 slides). The **study guide is the primary MCQ source**; the deck carries the **maths the guide omits**. Both are image-based scans, and the section numbers used in this reference are the guide's own.

Pair this with

The **In-Depth Workbook** for the same module — the same content taught from the ground up, with worked examples, lecturer's notes and a self-test at the end of every chapter. **This document is for recall; that one is for understanding.**